Tracking usage of ontologies in the translational science ecosystem

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Abstract

Translational research projects use ontologies to annotate, analyze, and harmonize diversely formatted data. Ontologies and the projects that use them are benefitted by ontology developers who are familiar with how their ontologies are used, but it is presently almost impossible for open-source ontology development teams to identify where their ontologies are being used outside their own laboratories. We propose to create a registry of ontology usage in translational research (ROUTR) to collect and disseminate information about how clinical translational science projects use and depend on ontologies.

Keywords

ontology, ontology usage, clinical translational science, database

1. Background

The biomedical sciences generate vast quantities of data. Ontologies are used to annotate and analyze these data and to harmonize information stored in diversely formatted knowledge bases [1, 2]. The integration of data from diverse sources has been identified as a persistent difficulty in translational science [3], a field that leverages interdisciplinary teams to create clinical innovations based on laboratory observations [4]. Accordingly, the integrative function of ontologies is of interest to translational research. Examples of the use of ontologies for this purpose include the standardization by the Human Immune Project Consortium of data from the immunology database ImmPort [5], the analysis and querying of MRI data by the University at Buffalo Center for Biomedical Imaging [6], and the prediction of the outcomes of cancer prevention-related behavioral change by the APRICOT Project [7]. These projects depend on a variety of terms from ontologies of molecules, cells, organisms, diseases, and scientific investigations.

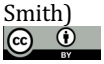
Many ontology browsers, like OntoBee and the Ontology Lookup Service, cross-reference between ontologies to serve information about which ontologies import terms from their source ontologies. These features exist because it is of value for an ontology's developers to know where its terms are being used. The ability to identify inter-ontology dependencies

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enables coordination of major changes between ontologies and motivates stability of the definition, axioms, and associated documentation of widely-used terms. However, these benefits are not exclusive to inter-ontology dependencies. When developers of a given ontology know that a database or other project depends on its terms, they will be able to make development decisions that are conducive to that usage, which can be cited as evidence of the value of the ontology, for example, in grant applications. Unlike inter-ontology dependencies, usage of ontologies by outside groups is opaque to ontology developers: there is no unified source of information about which projects depend on which ontologies. Furthermore, it has long been a challenge to document the usefulness of ontologies. Accordingly, we propose to create a registry of ontology usage in translational research (ROUTR) to collect and disseminate information about clinical translational science projects that use and depend on ontologies.

This registry will benefit ontology developers and users by enabling targeted notification of relevant changes to ontology terms. It may also allow the identification of the ways in which ontologies may (or may not) bring benefits to specific communities of users. While the use of ontologies continues to increase from year to year, there are still almost no robust data concerning any benefits conferred by ontology use. This registry may function as a step towards the collection of empirical data about the value of ontology usage in translational research.

2. Methods

In order to map dependencies between translational research projects and individual ontologies, the ROUTR must at minimum collect sufficient information to identify projects and the ontologies those projects use. The necessary data includes at least the name of the project and its principal investigator, a point of contact (if different from the PI), relevant institutional affiliations, and a funding source identifier (e.g., via grant numbers), plus a list of ontologies used by that project. Beyond this minimal base of information, it would ideally also include information about which ontology terms a project uses and how it uses those terms (for example through a list of terms to which a project maps values in its database). This secondary and more robust layer of information enables precise identification of which branches of a particular ontology are most widely used, which projects will be affected by modifications to a particular term, and how ontologies enable reasoning over data in practice. This data would be stored in a database and distributed in a freely accessible online knowledge resource.

Two possible strategies for collecting this information are web scraping and submission by the researchers who are involved in ontology-using projects. While web scraping is a more efficient means of collecting large amounts of data, user submission offers potential contextual insights that might be missed by web scraping alone. We propose leveraging both methods to develop as full a picture as possible of the use of ontologies in clinical translational research and the benefits of such use.

3. Relevance

We wish to present our proposal as a poster at ICBO 2026 because we believe the collection of data on usage of ontologies in translational science is relevant to the conference's third topic of interest, domain-specific applications in biomedical research. We aim to enable analysis of the benefits of practical implementation of ontologies in several of the domains named in that topic of interest, including biology, clinical & public health, health informatics, and behavioral science. Accordingly, this project is line with the conference's themes.

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